Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

Academic Year 2026-27

FY MCA- SEM I
IT12 Data Structure and Algorithms
Practical Assignment1

Date: 15/10/2026 **Marks: 40**
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Create a Singly Linear Linked List with following operations

1. Traverse and print the node values
2. Insert node at a specific position
3. Delete alternate nodes in the linked list

Code –

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class LinkedList:
    def __init__(self):
        self.head = None

    # 1. Create linked list (append at end)
    def append(self, data):
        new_node = Node(data)
        if self.head is None:
            self.head = new_node
            return
        temp = self.head
        while temp.next:
```

```
                temp = temp.next
        temp.next = new_node

    # 2. Traverse and print node values
    def display(self):
        if self.head is None:
            print("List is empty")
            return
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next
        print("None")

    # 3. Insert node at a specific position (1-based)
    def insert_at(self, data, pos):
        if pos < 1:
            print("Invalid position")
            return
        new_node = Node(data)
        if pos == 1:
            new_node.next = self.head
            self.head = new_node
            return
        temp = self.head
        for _ in range(pos - 2):
            if temp is None:
                break
            temp = temp.next
        if temp is None:
            print("Position out of range")
            return
```

```
                new_node.next = temp.next
            temp.next = new_node

        # 4. Delete alternate nodes (2nd, 4th, 6th, ...)
        def delete_alternate(self):
            temp = self.head
            while temp and temp.next:
                temp.next = temp.next.next
                temp = temp.next

ll = LinkedList()

n = int(input("How many nodes? "))
for i in range(n):
    ll.append(int(input(f"Enter value {i + 1}: ")))

print("\nOriginal list:")
ll.display()

val = int(input("\nEnter value to insert: "))
pos = int(input("Enter position: "))
ll.insert_at(val, pos)
print("After insertion:")
ll.display()

ll.delete_alternate()
print("\nAfter deleting alternate nodes:")
ll.display()
```

Output Screen

